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Mastitis in dairy cows: Diagnosis, management and economic impact

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Abstract
Mastitis remains a persistent and economically significant disease in dairy cattle, with subclinical mastitis (SCM) posing the greatest challenge due to its lack of visible symptoms and potential to impair milk yield and quality. In light of increasing concerns about antimicrobial resistance, many countries have implemented regulations restricting blanket dry cow therapy, promoting selective dry cow therapy (SDCT) to minimize unnecessary antibiotic use. This review explores key aspects of mastitis management, focusing on clinical signs, pathogens involved, diagnostic techniques, and treatment strategies. Bovine mastitis, whether clinical or subclinical, is mainly caused by bacterial pathogens such as *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Streptococcus dysgalactiae*, and results in elevated Somatic Cell Counts (SCC) and altered milk composition, particularly reduced casein and lactose levels. Early and accurate diagnosis is crucial for effective control. Conventional methods such as SCC measurement and bacteriological cultures remain gold standards, while novel technologies like metabolomics using GC-MS and NMR offer promising biomarker identification for early detection. On-farm diagnostic tools like the California Mastitis Test (CMT) and Easy Culture Systems facilitate rapid pathogen classification and treatment decisions. Treatment during the dry period is often more successful, especially for certain pathogens, while chronic cases often require culling. Preventive strategies, including proper hygiene, post-milking teat dipping, and environmental sanitation, are vital. This review emphasizes the importance of integrating traditional diagnostic methods with emerging technologies and responsible antibiotic stewardship. Future progress in mastitis control depends on advancing diagnostic tools, improving farm-level practices, and sustaining research efforts to reduce mastitis prevalence and its economic impact.

Keywords: Mastitis, subclinical mastitis, dairy cows, somatic cell count, selective dry cow therapy, diagnosis, metabolomics, milk quality, antibiotics, prevention strategies

Introduction
Mastitis continues to be among the most common ailments affecting the dairy industry, leading to significant financial losses worldwide. Among its forms, subclinical mastitis is the most widespread, causing decreased milk yield and increased somatic cell levels, and an increased risk of developing clinical mastitis during lactation, even in the absence of obvious symptoms. Due to rising public health concerns regarding antimicrobial use and its role in the development of antimicrobial resistance, many nations have implemented regulations aimed at limiting group-level prophylactic antimicrobial treatments. Selective dry cow therapy specifically targets antimicrobial treatment for cows or quarters suspected of intramammary infections, administering treatment after the last milking and sparing uninfected animals. Since the introduction of selective dry cow therapy, numerous methods for identifying infected cows or quarters have been reported.

This review discusses mastitis management in dairy cows and highlights the main diagnostic tools available, especially concentrating on tools used directly on farms. Mastitis is recognized as the most frequent and costly production disease among dairy herds in developed countries (Fourichon C. *et al*., 2001; Rajala-Schultz PJ) [9, 22].

Clinical Signs in Mastitis and Pathogens Involved
Bovine mastitis is generally categorized into clinical and subclinical types.

Clinical Mastitis (CM) is marked by local symptoms in the mammary gland and systemic symptoms such as redness, inflammation, pain, loss of appetite, elevated body temperature, reduced milk yield, and changes in milk composition (Peters *et al*., 2015) [19]. Severe cases may show abnormal milk, such as the presence of clots, watery secretions, or blood (Ferreira). In Subclinical Mastitis (SCM), there are no visible changes in the milk. However, increased somatic cell counts signal underlying inflammation. Additional indicators include elevated bacterial counts, reduced milk yield, and alterations in milk quality and composition (Bian, 2014) [4].

The primary causative agents of mastitis are bacteria, notably staphylococci and streptococci, along with mycoplasmas, fungi, algae, protozoa, and viruses. Mastitis may also arise from udder trauma, contact with chemical irritants triggering immune responses, or physiological injury (Shaheen and Tantary, 2015; Awale *et al*., 2012) [25, 3]. Among these, bacterial mastitis is especially significant due to its serious physiological effects and economic implications (Awale).

Diagnosis of mastitis

Early detection of mastitis and identification of its causative agents are crucial for effective control and treatment (Ashraf, A, Imran, 2018) [2]. These actions are essential for reducing treatment costs, minimizing milk yield losses, and improving cure rates (Miekley, 2013) [16].

Studies show that early treatment can significantly limit the severity of the disease and, in some cases, prevent visible symptoms (Milner, 1997) [17].

Diagnosis commonly relies on Somatic Cell Counts (SCC) and bacterial detection. Bacteriological culture and PCR analysis are considered the gold standards. New diagnostic tools like metabolomics are emerging. Metabolomics examines changes in metabolites to identify biomarkers for early, accurate detection. For example, gas chromatography-mass spectrometry (GC-MS) applied to blood identified valine, serine, tyrosine, and phenylalanine as prepartum markers and valine, isoleucine, serine, and proline as postpartum markers for subclinical mastitis (Hu *et al*., 2021) [11]. Nuclear Magnetic Resonance Metabolomics (NMR) on milk samples showed elevated levels of lactate, acetate, BHBA, butyrate, and isoleucine in high-SCC samples, with lower lactose, hippurate, and fumarate levels compared to low-SCC milk (Sundekilde).

On-farm culture systems like the Minnesota Easy Culture System II BiPlate and Tri-Plate help farmers make early treatment decisions by classifying pathogens into broad groups. These methods have higher specificity but slightly lower sensitivity compared to laboratory cultures.

The California Mastitis Test (CMT) detects major mastitis-causing pathogens with 82.4% sensitivity and 80.6% specificity during early lactation. For minor pathogens, sensitivity drops to 61%. At dry-off, CMT demonstrated 70% sensitivity and 48% specificity.

Effect on Milk Quality

Mastitis, whether clinical or subclinical, leads to a significant reduction in milk production and major changes in milk composition. Casein, the principal milk protein with high nutritional value, declines during mastitis, while lower-quality whey proteins increase, negatively impacting dairy product quality, particularly cheese production. According to Jones (2006) [13], increased Somatic Cell Counts (SCC) raise the levels of serum albumin and immunoglobulins in milk, reducing its heat stability and leading to lower grading after pasteurization. Mastitis also decreases calcium absorption from the blood into the milk, impairing milk coagulation properties, which are essential for cheese-making. Additionally, somatic cell-derived enzymes like plasmin degrade casein inside the udder before milk extraction. Mastitis alters milk mineral content-increasing sodium and chloride concentrations while reducing potassium levels. Since calcium is mostly bound to casein in milk, a reduction in casein lowers calcium availability as well. Reduced lactose concentrations in mastitic milk further impair acidification after starter cultures are added (Schallibaum, 2001) [24].

Milk from healthy quarters typically contains fewer than 200,000 somatic cells/mL. Elevated SCC to 300,000 cells/mL or higher indicates inflammation. During acute infections, SCC spikes within hours to days, while normalization after pathogen elimination may take weeks or longer (Jones, 2006) [13].

Treatment

Effective mastitis control focuses on sound husbandry practices, proper sanitation, post-milking teat dipping, antibiotic treatment during the non-lactating period, and culling of chronically infected animals. Therapeutic success is generally higher during the dry period than during lactation. Monitoring somatic cell counts and promptly diagnosing and treating infections is critical to mastitis management. *Staphylococcus aureus* remains the most challenging mastitis pathogen. Antibiotic cure rates during lactation are low, and many cows with chronic infections require culling. In contrast, *Streptococcus agalactiae* responds well to antibiotic treatment and can be eradicated from dairy herds with good mastitis control measures, including post-milking teat dipping and dry cow therapy. *Streptococcus dysgalactiae* survives in the udder, rumen, feces, and the barn environment and can be controlled with strict sanitation practices, showing moderate antibiotic susceptibility (Khan MZ and Khan A, 2006) [14].

Prevention strategies focus on minimizing bacterial exposure to the teat end by maintaining a clean, dry environment, ensuring cows avoid mud, manure, stagnant water, and dirty calving areas. Post-milking germicidal teat dipping is highly recommended. Attempts to use germicidal or barrier dips to control environmental mastitis during the dry period have been largely unsuccessful. Therefore, antibiotic therapy of all quarters at drying-off remains crucial for controlling environmental streptococci (Khan MZ and Khan A, 2006) [14].

Conclusion

Mastitis continues to pose major challenges for the dairy industry, severely impacting animal welfare and economic productivity. Effective management strategies emphasizing early detection, improved hygiene, selective antibiotic use, and culling of chronic cases are essential to controlling the disease. Technological advances such as metabolomics and on-farm culturing systems offer promising new diagnostic and control tools. Ongoing research, coupled with diligent implementation of best management practices at the farm level, is critical for reducing mastitis incidence and improving dairy herd health.

Conflict of Interest

Not available

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